

(a) (3 points) What is the maximum load that this bolt can support without deforming permanently?

(No plastic deformation, so $\sigma < \sigma_y$ | $\sigma_y = 350 (10^6) \text{ Pa}$) ①

$$\sigma_s = \frac{F}{A} \Rightarrow F = \sigma_y A \quad \left| \quad A = \left[\frac{\pi D_{\text{outer}}^2}{4} - \frac{\pi D_{\text{inner}}^2}{4} \right] \quad \begin{array}{l} D_{\text{outer}} = 10(10^{-3}) \text{ m} \\ D_{\text{inner}} = 5(10^{-3}) \text{ m} \end{array} \right.$$

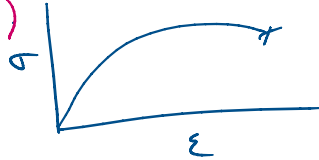
$$F_{\text{max}} = \frac{\pi \sigma_y}{4} (D_{\text{out}}^2 - D_{\text{in}}^2) = \frac{\pi \cdot 350(10^6)}{4} [(10(10^{-3}))^2 - (5(10^{-3}))^2] = 2.06(10^4) \text{ N}$$

with units ①

$F_{\text{max}} = 20 \text{ kN}$

(b) (3 points) What is the minimum load that will be required to fracture this bolt?

① Fracture, so need to exceed σ_{UTS} . Once necking begins Force required will decrease



$$F_{\text{min}} = \frac{\pi \cdot 420(10^6)}{4} (D_{\text{out}}^2 - D_{\text{in}}^2) \quad \text{or} \quad F_{\text{min}} = F_{\text{max}} \cdot \frac{420}{350} = 2.47(10^4) \text{ N}$$

with units ①

calculation ①

$F_{\text{min}} = 25 \text{ kN}$

(c) (4 points) The explosive is placed within the internal chamber and expands upon detonation, applying a pressure to the inner walls of the internal chamber. Assuming this pressure only acts to produce a tensile force against the circular ends of the internal chamber, what is the minimum pressure, in MPa, that must be applied by the explosive?

① $\phi 5 \text{ mm}$

① $P = \frac{F}{A}$ | $F_{\text{min}} = 25 \text{ kN}$ | $A = \frac{\pi D_{\text{in}}^2}{4}$ | $D_{\text{in}} = 5(10^{-3}) \text{ m}$ ①

① $P = \frac{4F}{\pi D_{\text{in}}^2}$

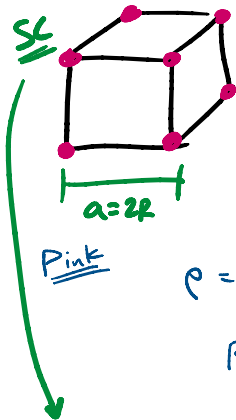
This concept sketch NOT necessary. ①

$$P = \frac{4 \cdot 25(10^3)}{\pi [5(10^{-3})]^2} = 1.26(10^9) \text{ Pa} \cdot \frac{1 \text{ MPa}}{10^6 \text{ Pa}} = 1.26(10^3) \text{ MPa}$$

with units ①

$P_{\text{min}} = 1300 \text{ MPa}$

2. (8 points) Some sources believe that, on his recent trip to space, William Shatner made an unscheduled spacewalk to collect some samples of several mysterious substances seen sticking to the outside of his New Shepard spacecraft. Upon his return from space he analysed the samples and tabulated all of the information he could obtain, shown in the table below. Using the data in the table, calculate the missing data for each mysterious substance. Assume each sample is a single element, with the exception of Chartreuse Matter, which is an ionic compound. Show your work.



Name	Radius (pm)	Crystal Structure	Molar Mass (g/mol)	Density (g/cm ³)
Pink Matter		FCC	81.4	8.96
Puce Matter	152	Simple Cubic		8.41
Periwinkle Matter	210	BCC	132.8	
Chartreuse Matter	$R_A = 152$ $R_C = :$	Rock Salt	$A_A = 101$ $A_C = 110$	4.98

Pink

$$\rho = \frac{nA}{V_c N_A} \quad \left\| \begin{array}{l} V_c = a^3 \\ n = 4 \end{array} \right\| \quad a = 2\sqrt{2}R \quad \left\| \begin{array}{l} \rho = 8.96 \text{ g/cm}^3 \\ A = 81.4 \text{ g/mol} \end{array} \right\|$$

$$\rho = \frac{nA}{(2\sqrt{2}R)^3 \cdot N_A} \Rightarrow R = \left(\frac{nA}{\rho N_A} \right)^{1/3} \cdot \frac{1}{2\sqrt{2}}$$

$$R = \left(\frac{4 \cdot 81.4}{8.96(10^{-30}) \cdot 6.022(10^{23})} \right)^{1/3} \cdot \frac{1}{2\sqrt{2}} = 138.6 \text{ pm} = \boxed{139 \text{ pm}} \quad \textcircled{1}$$

Puce

$$A = \frac{\rho V_c N_A}{n} \quad \left\| \begin{array}{l} V_c = a^3 \\ a = 2R \end{array} \right\| \quad A = \frac{\rho (2R)^3 \cdot N_A}{n} \quad \left\| \begin{array}{l} \rho = 8.41(10^{-30}) \text{ g/pm}^3 \\ n = 1 \end{array} \right\|$$

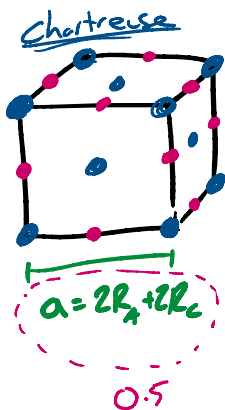
$$A = 8.41(10^{-30}) [2 \cdot 152]^3 \cdot 6.022(10^{23}) = 142.3 \text{ g/mol} = \boxed{142 \text{ g/mol}} \quad \textcircled{1}$$

Periwinkle

$$a = \frac{4}{\sqrt{3}} R \quad \left\| \begin{array}{l} \rho = 52.8 \text{ g/cm}^3 \\ A = 132.8 \text{ g/mol} \end{array} \right\|$$

$$\rho = \frac{nA}{\left(\frac{4}{\sqrt{3}} R \right)^3 \cdot N_A} \quad \left\| \begin{array}{l} n = 2 \\ R = 210 \text{ pm} \end{array} \right\|$$

$$\rho = \frac{2 \cdot 132.8}{\left(\frac{4}{\sqrt{3}} 210(10^{-10}) \right)^3 \cdot 6.022(10^{23})} = \boxed{1.54 \frac{\text{g}}{\text{cm}^3}} \quad \textcircled{1}$$



$$\rho = \frac{n_A A_A + n_C A_C}{(2R_A + 2R_C)^3 \cdot N_A} \Rightarrow R_C = \frac{\left(\frac{n_A A_A + n_C A_C}{N_A \cdot \rho} \right)^{1/3} - 2R_A}{2}$$

$$R_C = \left[\frac{(4 \cdot 101 + 4 \cdot 110)}{6.022(10^{23}) \cdot 4.98(10^{-30})} \right]^{1/3} - 2 \cdot 152 = 175.7 \text{ pm} = \boxed{176 \text{ pm}} \quad \textcircled{1}$$

$n_A = 4$
 $n_C = 4$
 $A_A = 101$
 $A_C = 110$
 $\rho = 4.98(10^{-30})$
 $R_A = 152$

3. (8 points) In the space below, describe how the strength of a metal can be increased by using each of the following crystalline imperfections: alloying elements, dislocations, grain boundaries, second phase particles.

All inhibit dislocation movement.

1) Alloying : ① (impurity atoms create lattice strain that either repels or pins dislocations.) $\uparrow$ alloying elements $\rightarrow$ $\uparrow$ strength.

2) Dislocations : ① (dislocations will either repel one another, or annihilate one another, making deformation harder locally.)
 $\uparrow$ dislocation density $\rightarrow$ $\uparrow$ strength.

3) Grain Boundaries : ① (dislocations must cross G.B.'s for plastic deformation.
 $\rightarrow$ change direction
 $\rightarrow$ lattice strain
 $\rightarrow$ planar mismatch. } require increased shear stress.
Since smaller grains have larger GB area per volume)
 $\downarrow$ Grain size $\rightarrow$ $\uparrow$ strength.

4) 2nd phase particles : ① (2nd phase particles typically hard : brittle $\rightarrow$ difficult for dislocation to move in, but must be able to pass particle for deformation.)

$\uparrow$ 2nd phase boundaries $\rightarrow$ $\uparrow$ strength.

This is the end of the written portion of the assessment.